Lesson plan · for the teacher

Reading numbers: units and scales

IoT · Level 1 · Lesson 2

Time
40 min

Materials
A classroom thermometer, or a paper thermometer with a sliding strip; A clear bottle of water with a paper scale taped on; Board and chalk; a ruler to draw a big scale; Worksheet: Reading detective (one per pair, optional)

Learning goals

- Read a thermometer scale at eye level, working out what each small step is worth.
- Record a reading with its number, unit, time and place.
- Explain why two readings must be taken the same way before we compare them.

Activity: Read the thermometer right: Reading round

1. Warm-up (5 min): Write “30” on the board. Ask: “Is 30 a lot?” Collect answers — 30 pupils, 30 minutes, 30 degrees? Then tell the story of two pupils who read 30 and 86 on one thermometer.
2. Teach (8 min): Write the four parts of a reading on the board: number, unit, time, place. Teach °C, mm and %. On the board scale, show how to find what each small step is worth.
3. Eye level (7 min): Hold up a clear bottle of water with a paper scale taped on. Pupils look at the water level from above, from below and level, and say what they notice. Agree the rule: eye level with the mark.
4. Reading round (14 min): Pairs read the board scale as you move a chalk line, writing full readings. Then they read the real or paper thermometer. If you have a real one, put it in the sun for five minutes and read again.
5. Wrap-up (6 min): Set up a class log by the door with columns for date, time, place and °C. Exit ticket: each pupil writes one full reading.

Hang the thermometer in the shade at pupils’ eye height an hour before class. Draw a big thermometer scale on the board from 0 to 50, with five steps between each labelled ten. Make a paper thermometer if the school has none. Tape a paper scale on a clear bottle of water for the eye-level demonstration.

Discussion

- What else do we need to know?
- Is it fair to compare the sun reading with the shade reading?
- What changed besides the weather?

Exit ticket

- Write one full reading: number, unit, time and place.

Try it in class tomorrow
Start a thermometer log: Hang a thermometer in the shade, at pupils’ eye height. Each morning at the same time, one pupil reads it at eye level and writes the number, unit, time and place in a class log. No thermometer? Draw a big scale on the board and practise.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **A thermometer's scale has five equal steps between 20 °C and 30 °C. What is each step worth?**

◯ 5 °C ◯ 2 °C ◯ 1 °C

2 **Which three entries have a problem in this pupil's weather log?**

☐ Monday: 24 °C, 8 am, shade by the door ☐ Wednesday: 35 °C, 1 pm, in the sun

☐ Thursday: 25 °C, 8 am, read from above ☐ Tuesday: 26

3 **You can read a scale from any angle if you look carefully.**

◯ True ◯ False

4 **Match each part of a reading to what it tells you.**

28 °C	• • • •	How full a phone battery is
12 mm	• • • •	When and where it was read
65 %	• • • •	How hot the air is
8 am, by the door	• • • •	How much rain fell

5 **Put the steps for reading the classroom thermometer in order.**

☐ Bring your eye level with the liquid

☐ Write number, unit, time and place

☐ Hang the thermometer in the shade

☐ Wait a few minutes for it to settle

Answers · for the teacher

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1 **2 °C**

30 − 20 = 10 degrees, shared over five equal steps, is 2 °C a step.

2 **Tuesday: 26; Wednesday: 35 °C, 1 pm, in the sun; Thursday: 25 °C, 8 am, read from above**

Tuesday has no unit, time or place: was it 26 °C at 8 am by the door?

3 **False**

Looking from above or below makes the reading seem higher or lower.

4 **28 °C — How hot the air is**
12 mm — How much rain fell
65 % — How full a phone battery is
8 am, by the door — When and where it was read

A full reading has four parts: a number, a unit, a time and a place.

5 **Top to bottom: 3, 4, 1, 2**

Shade first so the sun can't heat it, a short wait so it settles, eye level for an honest reading — then record all four parts.

For you to think about

Where could your pupils take a reading this week — and what unit, time and place would they write beside it?